LESSON 8.1

Understand Equations and Their Solutions

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can write an equation for a real situation and check whether a given value is a solution by substituting it.

WHY THIS MATTERS Detective Ruiz received a tip: 'A number of stolen gems plus 8 more equals 20 total gems.' She needs to write an equation to represent this clue before she can solve the case. How would you write this situation as an equation?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Equation *Spanish: Ecuación*	A math sentence with an equal sign showing both sides are the same.	x + 3 = 10
Variable *Spanish: Variable*	A letter that stands for a number that is unknown or can change.	the x in x + 3 = 10
Equal sign *Spanish: Signo igual*	The = sign, showing both sides are the same.	=
Expression *Spanish: Expresión*	A math phrase with numbers and letters, but no equal sign.	2x + 5 is an expression; it becomes an equation when you write 2x + 5 = 15
Constant *Spanish: Constante*	A fixed number in an expression or equation that does not change.	In x + 7 = 12, the numbers 7 and 12 are constants; only x can change.
Unknown *Spanish: Incógnita*	The value you are trying to find, usually shown with a variable.	In n + 5 = 12, n is the unknown; you solve to find n = 7.

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A basket has b apples. You add 9 more and have 15. Is b = 6 a solution of b + 9 = 15?

1. Substitute: replace b with 6 to get 6 + 9.
2. Add: 6 + 9 = 15.
3. Compare: 15 = 15, so the equation is true.

Answer: Yes, b = 6 is a solution.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

New clue: 'A number plus 5 equals 9.' What is the unknown?

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- I substituted _____ for the variable and got _____ .
- The value _____ (is / is not) a solution because _____ .
- An equation is true when both sides _____ .
- The variable _____ stands for _____ in this problem.

Word bank equation | variable | solution | substitute | true | false | equal sign | both sides | balance | check

WATCH OUT!

A common mistake

A common mistake in Write Equations is writing subtraction and division clues in the wrong order

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Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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1 **Circle the letter of the best answer. Show how you know.**

Which equation represents: 'A number plus 12 equals 30'?

(A) n + 12 = 30
(B) n − 12 = 30
(C) 12n = 30
(D) n / 12 = 30

STEPS TO TRY
1. Find the operation word in the sentence. What does 'plus' tell you to do with the number and 12?
2. Let n stand for the unknown number, then translate each phrase into symbols: 'a number' becomes n, 'plus 12' names an operation, 'equals 30' becomes = 30.
3. Build it left to right: n, then the symbol for 'plus', then 12, then '= 30'. Which choice matches that exact order?

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Which equation represents: 'Seven less than a number is 18'?

(A) n − 7 = 18

(B) 7 − n = 18

(C) n + 7 = 18

(D) n / 7 = 18

STEPS TO TRY

1. 'Seven less than a number' is tricky — the number comes FIRST, and then 7 is taken away from it.
2. 'Less than' flips the wording: '7 less than n' means start at n and subtract 7, not start at 7 and subtract n.
3. Write the number first: n, then subtract 7, then set it equal to 18. Which choice is built in that order?

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Which equation represents: 'Three times a number equals 21'?

(A) 3n = 21

(B) n + 3 = 21

(C) n − 3 = 21

(D) n / 3 = 21

STEPS TO TRY

1. Circle the operation word: 'times'. What operation does 'times' always signal?
2. 'Three times a number' means 3 multiplied by n, written 3n with no symbol between the 3 and the n.
3. Put 3n on the left and set it equal to 21. Which choice shows multiplication instead of adding or subtracting 3?

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Is n = 3 a solution of n + 5 = 9?

(A) Yes, because 3 is smaller than 9

(B) No, because 3 + 5 = 8, which does not equal 9

(C) Yes, because 3 + 5 is close to 9

(D) You cannot tell without solving first

STEPS TO TRY

1. Replace n with 3.
2. Work out the left side.
3. Does it equal the right side?

SHOW YOUR WORK

EXPLAIN YOUR THINKING

How did you decide what expression to put on each side of the balance?

SENTENCE STARTER I put ___ on the left side because ___, and ___ on the right side because ___. The equal sign means ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help